Lohit Gali— Senior Data Engineer
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PROFESSIONAL SUMMARY:

- Results-driven Senior Data Engineer with over 5 years of comprehensive experience in designing and implementing robust data solutions.
- Profound expertise in architecting and optimizing scalable data pipelines across diverse cloud environments including Azure and AWS.
- Demonstrated proficiency in advanced Big Data technologies such as Spark (Scala/PySpark), Databricks, Snowflake, and Kafka for large-scale data processing.
- Specialized in leveraging Azure cloud components like ADLS Gen2, Azure Data Factory, Azure Synapse Analytics, and Azure Databricks for enterprise data solutions.
- Skilled in developing complex ETL/ELT processes, ensuring data quality, and performing performance tuning for critical data platforms.
- Adept at designing and implementing real-time streaming solutions using Kafka and Azure Stream Analytics for immediate data insights.
- Experienced in data modeling, database management with SQL, NoSQL, Oracle, and PostgreSQL, and advanced data warehousing practices.
- Proven ability to implement CI/CD pipelines, containerization with Docker, and automated deployment strategies for data applications.
- Strong commitment to data governance, security protocols, and compliance standards within highly regulated industries.
- Excellent communicator and collaborative team player, consistently driving project success through technical leadership and problem-solving.

TECHNICAL SKILLS:

- **Programming Languages:** Python, Scala, SQL, PySpark
- **Cloud Platforms:** Azure (ADLS Gen2, Data Factory, Synapse Analytics, Databricks, Stream Analytics, DevOps, Monitor, Cosmos DB), AWS (S3, EMR, Glue, Redshift, Kinesis, Lambda, Athena, EC2, IAM), Google Cloud Platform
- **Big Data Technologies:** Databricks, Spark, Snowflake, Kafka, Hadoop, Hive, Delta Lake
- **Database Management:** Azure SQL Database, PostgreSQL, Oracle, MySQL, MongoDB, MS SQL SERVER
- **Orchestration & CI/CD:** Azure Data Factory, Airflow, Jenkins, Docker, Kubernetes, Azure DevOps, Git, GitLab
- **Data Warehousing & BI:** Snowflake, Azure Synapse Analytics, Tableau, Power BI
- **Other Tools & Methodologies:** Confluence, JIRA, ServiceNow, Lucidchart, Agile, Scrum, ETL Design, Data Modeling

WORK EXPERIENCE:

Senior Data Engineer @ UnitedHealth Group — Hartford, CT **June 2023 – Present**

- Architected and implemented highly scalable data pipelines using Azure Data Factory and Azure Databricks for claims and patient data.
- Designed and managed Azure Data Lake Storage Gen2 (ADLS Gen2) for ingesting and storing vast volumes of structured and semi-structured healthcare data.
- Developed PySpark transformations within Azure Databricks to cleanse, enrich, and aggregate complex datasets for analytical consumption.
- Engineered robust data ingestion frameworks leveraging Azure Data Factory to pull data from various sources into ADLS Gen2, including Oracle.
- Implemented real-time streaming solutions using Azure Stream Analytics and Kafka to process critical healthcare events with minimal latency.
- Optimized data warehousing solutions within Azure Synapse Analytics, creating external tables and enhancing query performance for reporting.
- Ensured strict data quality and governance standards by implementing advanced validation rules and monitoring through Azure Monitor.
- Migrated legacy on-premise data systems to Azure cloud, focusing on cost optimization and performance improvements within the Azure ecosystem.
- Collaborated with data scientists to integrate machine learning models into data pipelines, deploying them efficiently via Azure DevOps.
- Established robust CI/CD pipelines using Azure DevOps to automate testing, deployment, and version control for all data engineering assets.

- Developed comprehensive documentation for data architecture, pipeline flows, and technical specifications using Confluence and Lucidchart.
- Participated actively in Agile Scrum ceremonies, managing tasks and backlog items efficiently using JIRA for project tracking.

Technologies Used: Azure (ADLS Gen2, Data Factory, Databricks, Synapse Analytics, Stream Analytics, Monitor, DevOps), PySpark, Kafka, Snowflake, Oracle, JIRA, Confluence, Git

Data Engineer @ Liberty Mutual Insurance — Boston, MA **July 2021 – August 2022**

- Designed and built end-to-end data pipelines on Azure to process large-scale policy, claims, and underwriting datasets efficiently.
- Developed robust ingestion frameworks utilizing Azure Data Factory to extract data from PostgreSQL and other sources into Azure Data Lake Storage Gen2.
- Implemented advanced PySpark transformations on Azure Databricks for comprehensive cleansing, standardization, and aggregation of insurance data.
- Engineered real-time data streaming pipelines using Kafka for capturing and processing critical events with low latency for immediate insights.
- Optimized data warehousing performance in Azure Synapse Analytics by creating external tables and refining SQL queries for complex analytical reports.
- Developed and maintained Delta Lake tables for efficient incremental data loading and robust data versioning, ensuring data consistency.
- Managed metadata-driven ingestion patterns to facilitate the seamless onboarding of new data sources and improve pipeline reusability.
- Implemented containerized data processing solutions using Docker to ensure consistent environments across development and production.
- Orchestrated complex data workflows using Control-M, integrating various Azure services and external data systems effectively.
- Enforced stringent data security measures with role-based access control (RBAC) and encryption protocols within the Azure environment.
- Authored detailed architectural designs and data flow diagrams using Confluence, ensuring clear documentation of data processes.
- Supported Agile delivery methodologies, tracked progress using ServiceNow and JIRA, and actively participated in sprint planning.

Technologies Used: Azure (ADLS Gen2, Data Factory, Databricks, Synapse Analytics), PySpark, Kafka, PostgreSQL, Delta Lake, Docker, Control-M, Git, Confluence, JIRA

Data Engineer @ Target — Minneapolis, MN **July 2019 – June 2021**

- Developed and maintained scalable ETL pipelines using AWS Glue and EMR to process vast POS, product, and customer datasets for retail analytics.
- Engineered data ingestion workflows from Oracle and other relational databases into Amazon S3 using AWS Glue crawlers and custom scripts.
- Designed and implemented batch processing pipelines using Spark-Scala on AWS EMR clusters for cleansing and standardizing large retail data volumes.
- Created and optimized Hive tables within AWS EMR, leveraging partitioning and bucketing strategies to enhance query performance for reporting teams.
- Supported the migration of on-premise Teradata systems to AWS Redshift, building new data consumption layers and optimizing data structures.
- Implemented robust workflow orchestration using Apache Airflow on AWS EC2 instances for scheduling and managing complex data dependencies.
- Performed advanced SQL analysis within AWS Athena and Redshift for marketing, sales, and inventory teams, providing critical business insights.
- Developed efficient data storage solutions utilizing Parquet and Avro formats on Amazon S3 for improved analytical performance and cost efficiency.
- Applied comprehensive data quality rules and validation frameworks for validating product master and store master data effectively.
- Implemented custom logging frameworks using Log4j for enhanced monitoring and debugging of Spark applications on AWS EMR.

- Participated actively in Agile sprints, documented data transformations and technical specifications within Confluence and SharePoint.
- Provided crucial support for production deployments and managed version control of data assets using Git and GitLab repositories.

Technologies Used: AWS (S3, EMR, Glue, Redshift, Athena, EC2), Spark-Scala, Hive, Airflow, Oracle, PostgreSQL, Git, Confluence, SharePoint

EDUCATION:

Master of Science in Computer Science @ Oklahoma State University